RealQM for Molecules: Validation

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Mathematical formulation by the author; implementation and numerical experiments developed with AI assistance (Claude, Anthropic).

Abstract

RealQM [1] describes a molecule as a partition of space into non-overlapping domains, each holding one electron (or a pair counted once) as a real density, with the domain boundaries placed where the density is continuous and each electron free at its wall (a homogeneous Neumann condition). We solve this *hard-partition* model as it stands, converged in the mesh, with nothing adjusted to molecular data (each core radius is set once per atom, from its ionisation energies), using a three-dimensional solver (`molecule.js`) and, for linear molecules whose boundaries are fixed by symmetry or construction (homonuclear X_2, where the boundary is the midplane), an exact axisymmetric reduction that serves as an independent check: on H_2 the two give binding energies of 0.2251 and 0.2241 E_h at the same mesh. Each electron pair can be carried in two parameter-free constructions to choose from: *2split*, two one-electron domains (complete, permanent avoidance), and *2hom*, one pair domain (no avoidance beyond the mean field, the Hartree–Fock level). Both are reported; for H_2 and He the exact value lies between them. Where the boundary is fixed by symmetry the bond is sharp: H_2 -2.3%, converged. A free boundary carries no Pauli cost, so inner shells are excluded by a fixed core radius per atom: every all-electron LiH with free core boundaries collapses, while the pseudo-Li core with one 2hom bonding pair gives $R_e = 2.85\,a_0$ (exact 3.015), $D_e = 0.095\,E_h$ (exact 0.092) and a dipole of the right sign, 11% high. The model is judged by geometry. Where an atom carries many valence electrons its binding carries a large offset that depends on the composition and hardly on the geometry—CH_4 over-binds by $3.2\,E_h$, constant to $0.03\,E_h$ over $R = 2.05$–2.45, yet its bond comes out 10% long and the tetrahedron is preferred to the square—so atomisation energies are a diagnostic, and energies are quoted only for processes in which every atom keeps its bonding. Free angles are not yet in range: the bond angle of H_2O closes below 95°. Molecules are built by successive reduction—all electrons, frozen $1s^2$ core, reduced dimension for X_2, reduced valence charges on a coarse mesh—and the last step carries proteins of over two thousand atoms on one GPU in a web browser.

1 The model

RealQM [1] replaces the many-electron wave function by one-electron densities on non-overlapping domains of ordinary three-dimensional space. A molecule with nuclei $\{(\mathbf{R}_a, Z_a)\}$ is described by domains Ω_k partitioning space, each carrying $n_k \in \{1, 2\}$ electrons (two only of opposite spin) as a real amplitude u_k with $\int_{\Omega_k} u_k^2 = 1$ and density $\rho_k = n_k u_k^2$. The energy is

$$E = \sum_k n_k \Big(\tfrac{1}{2} \int_{\Omega_k} |\nabla u_k|^2 - \int_{\Omega_k} K u_k^2 \Big) + \sum_k \tfrac{n_k(n_k-1)}{2} U_{kk} + \sum_{k<l} n_k n_l U_{kl} + V_{KK}, \tag{1}$$

with $K = \sum_a Z_a/|\mathbf{x} - \mathbf{R}_a|$, U_{kl} the Coulomb interaction of the unit densities u_k^2 and u_l^2, and V_{KK} the nuclear repulsion. A pair domain ($n_k = 2$) counts its single pair once.

Boundary conditions. Each u_k satisfies a homogeneous Neumann condition on $\partial\Omega_k$: the kinetic energy is counted only inside the domain, and the amplitude at the wall is free. The boundaries are placed by *density continuity*, $\rho_k = \rho_l$ across each interface. Energy minimisation over the boundary position is not used: in this model a ragged or displaced boundary can lower the energy (Section 5.3), and only continuity holds it.

2 Atoms

Atoms are computed in spherical symmetry, with all electrons organised in non-overlapping radial shells following the Atom Simulator configurations [2] (Li 2+1, B 2+3, ...): a one-dimensional problem in the radius, with each shell boundary free and placed by the same continuity rule as in three dimensions. For use in molecules the atom is then reduced to a *pseudo-atom*: the $1s^2$ core (and further inner shells) is frozen into a pseudo-nucleus of charge Z_x with a hard core of radius r_c, and only the outer shell is carried in three dimensions, split into domains of at most two electrons (boron: pseudo-B +3 with an outer 2hom pair and one electron). A radial shell holding three or four electrons is a shared density in one dimension, which no domain may be in three dimensions; it stands for the spherical average of its angular split into domains of at most two (boron's 3 as 2_{hom}+1, a 4 as a tetrahedron of single electrons). Shells hold at most four—the pair and the tetrahedron, an octet being 4+4, never a single shell of six or eight. Energy does not select the configuration: more sharing always lowers the one-dimensional energy (Li as one shell of 3 is 13.7% too deep, as 2+1 within 0.5%), so the configuration is set by the Pauli cap of two per domain, not by minimisation. The core radius is a property of the atom, not of any molecule, and it is obtained in two ways. *Fitted*: the radius at which the pseudo-atom alone, in three dimensions, reproduces the sum of the ionisation energies of the electrons it carries. *Computed*: the core–valence interface of the one-dimensional all-electron atom, which for a $1s^2$ core is fixed by the model itself, independent of the box:

	r_c fitted to IE (3D)	r_c computed (1D, boxes 6–20 a_0)	difference
Li (2+1)	1.99	1.98–2.05	within 3%
B (2+3)	1.08	0.92–1.00	8–17%
Be (2+2)	1.37	pending	
C (2+4)	1.0	pending	

For Li the two agree, and the core radius is an output of the model. For B the computed radius is shorter than the fitted one. We use the fitted radius throughout, and state the computed one beside it; an outer core shell beyond $1s^2$ (Na and below) has no box-independent interface in one dimension, and there only the fitted radius exists. At $h = 0.1$ the hard core excludes whole cells, so the fitted radius is defined to a few hundredths of a_0.

3 Molecules by successive reduction

Molecules are built by reducing the full model step by step, each step tested on systems where the step before it can still be run.

1. *All electrons* (H_2, He): every electron in its own domain or in a pair, with no pseudo-nucleus. Here an electron pair has two constructions to choose from, 2split and 2hom (Section 5).

2. *Frozen* $1s^2$ *core*: for every molecule beyond H_2 the atoms enter as pseudo-atoms (Section 2); only valence electrons are computed, in domains of at most two, with the inner boundary r_c fixed and all outer boundaries free. A free boundary carries no Pauli cost, so the core sphere must be fixed (Section 6.1). LiH and BH are computed this way.

3. *Reduced dimension* for linear molecules whose boundaries are fixed planes—the midplane of a homonuclear X_2, by symmetry, or planes through the nuclei, by construction: the axisymmetric solver (Section 4).

4. *Large molecules*: pseudo-atoms with reduced valence charges on a coarse mesh. The same `molecule.js` then carries a 238-residue protein (green fluorescent protein, 2307 atoms) on a 300^3 grid at $h = 0.4\,a_0$, and crambin (540 atoms) on 200^3 at $h = 0.25\,a_0$, on the GPU of a laptop in a web browser, in minutes to hours. At this level the mesh no longer resolves the bond, and the protein folds shown are guided by contact restraints; the step makes the computation possible, and what it preserves is tested only on small fragments (Section 7).

4 Solvers, and the check between them

`molecule.js`. One amplitude field on a Cartesian grid, each cell labelled by its owning domain, so the domains are disjoint by construction. The kinetic stencil omits every face joining two labels (true Neumann). Boundaries move by label flips: a boundary cell changes owner when the neighbouring domain's density exceeds its own. The Coulomb potential of the other electrons is a direct per-electron Poisson solve. The nuclear kernel is clamped, $-Z/\max(r, R_{\text{sing}})$, with $R_{\text{sing}} = 2h$ unless pinned. The amplitude is relaxed in imaginary time; convergence is declared when the mean energy over a window of 3000 steps changes by less than $10^{-4}\,E_h$ between successive windows, since a free boundary keeps flipping single cells indefinitely and a step-to-step test either never fires or fires early.

Axisymmetric reduction. For a homonuclear X_2 the domain boundary is the midplane by symmetry, and the model reduces exactly to (z, ρ) with a plane boundary. The same holds for any linear molecule whose walls are all fixed planes, such as planes through the nuclei separating the halves of a pseudo-atom's valence, with every bond a 2hom pair (Section 6.3). We use it only there: a free wall between unlike atoms curves, and a plane would impose it. The Neumann condition is imposed by assembling the stiffness only over faces inside a domain. Each domain's lowest state is an exact sparse eigensolve, and the potentials are iterated to self-consistency. Mesh ladders down to $h = 0.05$ are cheap.

The check. The electron–electron energy of the two solvers agrees to 1.4%, and the total energies to the extent their kernels allow. At $R = 2.0\,a_0$, $h = 0.1$ and box $10\,a_0$:

	$E(H_2)$	$2E(H) - E(H_2)$
`molecule.js` (3D, free boundary)	−1.20734	0.2251
axisymmetric (2D, exact eigensolve)	−1.23685	0.2241
exact	−1.1381	0.1381

The totals differ by the kernel (clamped at $2h$ in 3D, analytic in 2D), which largely cancels in the binding energy; the binding energies agree to 0.5%. The three-dimensional solver reproduces the model.

5 Hydrogen and helium: two constructions of a pair

5.1 Two constructions of an electron pair

RealQM carries no spin variable; spin enters only through the Pauli principle, as the rule that a domain holds at most two electrons, and two only if of opposite spin. Electrons in different domains are excluded from each other by disjointness, whatever their spin. A pair of opposite spin can therefore be carried in two parameter-free ways. In the *2split* construction each electron owns its own domain and the two meet at a free wall: their avoidance is complete and permanent. In the *2hom* construction both electrons share one domain and one density, the overlay that Pauli permits for opposite spins, each feeling half its Hartree field: there is no avoidance beyond the mean field, and the pair is a restricted Hartree–Fock pair. A one-dimensional radial check on Li^+ makes the last point exact: the 2hom $1s^2$ pair gives -7.23630 against the Hartree–Fock -7.2364. The true state avoids partially and dynamically.

Why two electrons may share a domain, and not three, is a statement about the kinetic energy. Read as the self-repulsion of a charge cloud, a pressure against compression, the kinetic energy of a density ρ is the von Weizsäcker term $T_{\mathrm{vW}}[\rho] = \frac{1}{2}\int |\nabla\sqrt{\rho}|^2$. For a 2hom pair, $\rho = 2u^2$ and $T_{\mathrm{vW}}[2u^2] = 2 \cdot \frac{1}{2}\int |\nabla u|^2$: the pair is one cloud of charge two, its compression energy that of the combined density, and its mutual repulsion counted once. In standard quantum mechanics the same holds—for one or two electrons of opposite spin in one orbital the kinetic energy is exactly T_{vW} of the density—so within a domain the 2hom pair coincides with the restricted Hartree–Fock pair. For three or more, standard quantum mechanics adds the Pauli kinetic energy $T_{\mathrm{P}} = T_s - T_{\mathrm{vW}} > 0$, the cost of the nodes that antisymmetry forces between electrons of the same spin. A domain holding three would carry only T_{vW} and collapse; the third electron therefore takes its own domain, and disjointness takes over the exclusion. Two is the number of electrons whose kinetic energy the shared density describes exactly.

5.2 H_2 and He

For both two-electron systems the exact value lies between the two constructions (H_2 axisymmetric, $R = 1.4$, $h = 0.1$):

H_2	T	V_{eK}	V_{ee}	D_e
2split, two one-electron domains	1.291	−3.843	0.545	0.281
exact	1.1745	−3.650	0.587	0.1745
2hom, one pair domain	1.153	−3.679	0.664	0.138

He	E
2split (two hemispheres)[a]	−3.026
exact	−2.9037
2hom (one pair, 1D radial)[b]	−2.845

[a] Axisymmetric; the pair repulsion is 4% below exact. [b] The Atoms paper [2]. (For H_2 the exact T is $-E$ by the virial theorem, and V_{eK} follows.) In both cases the exact value lies strictly between the two constructions, with no adjustable quantity on either side: $D_{2\mathrm{hom}} < D_e < D_{2\mathrm{split}}$ and $E_{2\mathrm{split}} < E < E_{2\mathrm{hom}}$. This is the model's energy statement. An intermediate construction that reproduced the exact value would be a fit.

The two constructions are distinct cases with no bridge between them. A pair either has a wall through it or it does not, and no parameter of the model carries one into the other. Charging the wall (Section 5.3) does not do it: as β grows the density moves away from the wall, not towards the

shared pair. The situation is like a bifurcation. The pair sits on one branch or the other, and the states in between are not solutions of the model. The exact value lies between the two branches in energy, but it is not on a path that joins them.

The geometry is sharper than the energy. The 2split H_2 converges in the axisymmetric reduction to $R_e = 1.368\,a_0$, short of the exact 1.401 by 2.3% and flat to 0.2% under a 2.5-fold mesh refinement.

What makes the H_2 bond is not agreed among experts, even though the exact energy is not in doubt. One account, from Slater [7] and Feynman [8], is electrostatic: charge gathers between the nuclei, and the Hellmann–Feynman force on each nucleus pulls it inward. The other, from Hellmann [9] and Ruedenberg [10], puts the cause in the kinetic energy: an electron spread over both atoms lowers its kinetic energy, and the potential energy follows. The two accounts are still argued today [11]. Both read the same virial balance, $\Delta T = -\Delta E$ and $\Delta V = 2\Delta E$ at equilibrium, which holds whichever is the cause. The model takes no side by assumption. Its 2split bond is made by charge held in the bond region at no kinetic cost (Section 5.3), so it is closer to the electrostatic account. That is a property of the construction, not a settlement of the question.

5.3 What places the 2split side

The 2split energy is not an error in the solution: two independent solvers agree, the energy converges, and every term is counted. Its components show what the free wall does. Against exact, the nuclear attraction is larger (centroid of electron 1 at -0.785, the nucleus at -0.70) and V_{ee} smaller: a Neumann wall lets each electron keep full amplitude right up to the midplane—in the bond region, where both nuclei attract it—at no kinetic cost, and the other electron, confined strictly to the other side, never competes for that region. Charging the wall shows this directly. With a surface energy $\frac{\beta}{2}\int_{\partial\Omega_k} u_k^2$ on the interface:

β	T	V_{eK}	V_{ee}	centroid
0	1.291	−3.843	0.545	−0.785
0.15	1.277	−3.803	0.526	−0.832
0.5	1.265	−3.695	0.485	−0.936
2	1.281	−3.255	0.381	−1.269
10	1.238	−2.607	0.288	−1.713

As soon as the wall costs anything the density leaves the interface for the far side of its own nucleus and the nuclear attraction falls, with the kinetic energy almost unchanged. The free cut at the wall is the model's central assumption; it sets the 2split energy, while the 2hom pair has no wall.

5.4 Two states of a pair

The two constructions are two different states of the pair, and standard quantum mechanics has both. The 2hom pair, two electrons of opposite spin carrying one density over one domain, is the restricted Hartree–Fock singlet: one doubly occupied orbital. In H_2 at $R = 1.4$ the axisymmetric 2hom energy is -1.133585 at $h = 0.1$, against the Hartree–Fock limit -1.133630 [12]. The two share the known Hartree–Fock behaviour: H^- is unbound, and the pair does not dissociate into two neutral atoms, because half of it stays ionic. The 2split pair is the Heitler–London state. Each electron keeps to its own region, which is avoidance in the extreme, and spin plays no part: without an exchange term the singlet and the triplet are the same. It dissociates correctly and binds H^-. The exact ground state is neither. It is a correlated singlet, which standard quantum mechanics builds by mixing the shared and the separated configurations. A real density with one construction

per pair has no such mixture, and that is why the exact energy lies between the two and is reached by neither (Section 5).

In Hartree–Fock the two states are joined by a bifurcation. The restricted solution is stable near equilibrium and becomes unstable beyond the Coulson–Fischer point, about $R = 2.3$, where it breaks symmetry into the unrestricted solution, which separates the electrons like 2split [13]. The model has the two branches but not the path between them. Domains are disjoint, and two electrons share a domain only with the same density, so the partial overlap through which the Hartree–Fock solution breaks symmetry is not available. A 2hom pair therefore cannot decay into a 2split pair. It is also lower in energy at no R:

R	D(2hom)	D(2split)	$E_{\text{split}} - E_{\text{hom}}$
1.0	0.0847	0.2305	−0.1458
1.2	0.1246	0.2679	−0.1434
1.4	0.1332	0.2747	−0.1416
1.6	0.1259	0.2663	−0.1404
2.0	0.0912	0.2308	−0.1396
2.4	0.0489	0.1894	−0.1406
3.0	−0.0111	0.1336	−0.1447
4.0	−0.0887	0.0679	−0.1566
5.0	−0.1410	0.0308	−0.1718
6.0	−0.1755	0.0114	−0.1869
8.0	−0.2135	−0.0037	−0.2098

(Axisymmetric, $h = 0.1$, $D = 2E_{\text{H}} - E$ on the same mesh; 2hom mirror-symmetric.) The two curves do not cross: 2split is lower at every R, by 0.14 near equilibrium and by more as the bond stretches. If a path existed, energy minimisation would take every pair to 2split, including a bonding pair at equilibrium, where 2hom is the closer one to experiment (Section 5.5). The free wall over-favours avoidance. So the construction of a pair is not chosen by energy. It is chosen by the physical rule of the next subsection: shared pairs are 2hom, and electrons that must avoid each other are 2split.

Breaking a bond requires 2split. The scan shows the division of labour directly. The 2hom pair stops binding near $R = 3$ and rises steeply beyond it, 0.21 above two atoms at $R = 8$, because half of the shared pair stays ionic, as in restricted Hartree–Fock. Left without the mirror symmetry, a stretched 2hom pair does not turn into a 2split pair. It collapses onto one nucleus, the ionic state $\text{H}^- + \text{H}^+$ (97.5% of the density on one side at $R = 6$), which is a higher solution than the symmetric pair. Only 2split goes to two neutral atoms. Its tail falls off too slowly, however: it still binds 0.011 at $R = 6$, where the exact binding is nearly zero, and it is slightly repulsive (0.004) at $R = 8$. This matters for the later aim of reactions between fragments. A reaction breaks or forms bonds, so somewhere along its path a pair has to change from 2hom to 2split, and the model has no path between them. The energies at the two ends of a reaction can be computed in the model; a barrier between them needs a rule for where the pair changes, which the model does not yet have.

A bridge by competition. Reaction–diffusion systems suggest a bridge. Two densities that diffuse and compete through a term $\kappa u_1^2 u_2^2$ separate into disjoint regions as $\kappa \to \infty$, the strong-competition limit [14]. We give each electron its own density over the whole space and add $\kappa \int u_1^2 u_2^2$ to the energy. At $\kappa = 0$ this is unrestricted Hartree–Fock for two opposite spins: the Hartree repulsion between two orbitals, with no exchange term. Started each time from the electrons on opposite sides (axisymmetric, $h = 0.1$):

R	κ	D	orbital overlap	state
1.4	0	0.1332	1.000	symmetric = 2hom
1.4	5	−0.0429	1.000	symmetric
1.4	6	−0.0707	0.973	broken
1.4	1000	−0.5604	0.045	segregated
2.2	0	0.0702	1.000	symmetric
2.4	0	0.0497	0.923	broken
4.0	0	0.0012	0.265	broken

The family has a pitchfork bifurcation. At $\kappa = 0$ the symmetric solution breaks between $R = 2.2$ and 2.4, which is the Coulson–Fischer point. At $R = 1.4$ it breaks between $\kappa = 5$ and 6. Near both breaks the iteration slows down, as it does at a bifurcation. In the (R, κ) plane a curve of symmetry breaking separates the shared pair from separated electrons, and the broken $\kappa = 0$ branch dissociates correctly ($D = 0.001$ at $R = 4$). The far end of the family, however, is not 2split. Segregation by competition forces the densities to vanish at the interface, and this hard wall costs energy: at $R = 1.4$, $\kappa = 1000$ is 0.56 above two atoms, while 2split binds by 0.275. No overlap penalty reaches the free Neumann wall. The wall is the model's own assumption and the source of its over-binding (Section 5.3), and 2split stands outside the family of overlapping states. A switching rule by competition is therefore available, but it switches into the hard-wall branch. Using it for reactions would add these states to the model rather than join its two constructions.

5.5 Beyond two electrons

The same two constructions apply to any pair in a molecule. For LiH, with Li as a +1 pseudo-nucleus ($r_c = 1.99$ from its ionisation energy) and the bonding pair as 2split (Li valence and H electron in separate domains) or 2hom (one pair over Li^+ and a bare proton, Section 6.1), D_e is 0.063–0.073 (2split) and 0.095 (2hom) against the exact 0.092. For BH (pseudo-B +3 with a 2hom lone pair) D_e is 0.276 (2split) and 0.155 (2hom) against 0.135 (Table 1). In both, the 2hom bonding pair is the closer.

The picture also tells where each construction belongs. A pair that shares one region—a core $1s^2$, or a bonding pair near equilibrium—is naturally 2hom. Electrons that must avoid each other need 2split: the 2hom H^- is unbound, exactly as in Hartree–Fock, while the 2split H^- binds; and a 2hom pair cannot dissociate a bond into two neutral atoms. We report both, and choose by this rule.

The results below divide the roles as follows. *Geometry is common to both cases:* where both were run, the bond lengths differ by a few per cent at most (Li_2 5.1/5.15, N_2 1.95/1.95, LiH 2.85/2.98, BH 2.29/2.38; Table 1 and Section 6.3), so the bond length does not depend on the choice. *2split is the general case and over-binds:* its D_e is above experiment in most molecules computed, up to about twice (BH 0.276 against 0.135, OH 0.371 against 0.170). The exceptions are LiH (0.063–0.073 against 0.092) and B_2 (at experiment). *2hom is the Pauli pair:* two opposite spins sharing one domain, the most Pauli exclusion allows, with kinetic energy exactly T_{vW}. It is the closer energy for a shared pair at equilibrium (LiH 0.095 against 0.092, C_2 0.242 against 0.232, N_2 0.370 against 0.364) and under-binds where the electrons should avoid each other. It also gives the LiH dipole, of the right sign and 11% high, where one electron per domain inverts it. Dipoles in general are not settled by either case. In BH only a cone-shaped boundary corrects the dipole, at the cost of bond length and binding (Section 6.2). In OH both cases are about twice too large. H_2O needs split O for the right sign.

6 Results

The criterion is geometry: bond lengths, and where tested the preferred shape. Binding energies are listed as a diagnostic of the construction, not as a target; across compositions they carry an offset that cancels only within one composition (Section 6.4). **[In progress.]** Bond lengths from an R-scan at three meshes ($h = 0.10, 0.07, 0.05$) per bond with `molecule.js`, extrapolated in h, with the kernel clamp pinned so that the ladder refines a fixed problem; axisymmetric values beside them for X_2 only. Energies for both constructions, 2hom and 2split, at the computed geometry.

		bond length (a_0)				binding D_e (E_h), diagnostic		
molecule	construction	3D	axisym	exp.	error	2hom	2split	exp.
H_2	1+1	pending	1.368	1.401	−2.3%	0.138	0.281	0.1745
LiH	Li^+ (r_c) + pair	2.85	—	3.015	−5%	0.095	0.063–0.073	0.092
BH	B 2+3 (2_{hom}+1)	2.29	—	2.329	−1.6%	0.155	0.276	0.135
NH	N 2+2+3	pending	—					
OH	O +3 (2_{hom}+1)	1.79	—	1.832	−2%	0.241	0.371	0.170
BeH_2	Be $2+2_{\mathrm{split}}$	2.44	—	2.52[a]	−3%	0.248[b]	n.c.	0.2231
HF	F 2+4+3	pending	—					
NaH	Na	pending	—					
H_2O	O +3 (1+1+1)	obs.[c]	—	1.809	—	0.651	n.c.	0.37
NH_3	N +3 / +5	obs.[c]	—	1.913	—	1.91 / 3.45	—	0.47
CH_4	C +4 (1+1+1+1)	2.25[d]	—	2.054	+10%	3.83	3.82	0.67

Table 1: Bond lengths and the 2hom and 2split binding energies against experiment. H_2 energies are axisymmetric ($R = 1.4$, $h = 0.1$); LiH is three-dimensional (box $20\,a_0$, $h = 0.1$), its bond length from the 2hom construction (2split: 2.98–2.99). BH is three-dimensional (box $12\,a_0$, $h = 0.1$, pseudo-B $r_c = 1.08$), $R = 2.1, 2.3, 2.5$; bond length from 2hom (2split: 2.38). BeH_2: pseudo-Be $r_c = 1.37$, box $12\,a_0$, $R = 2.3, 2.5, 2.7$; [a]gas-phase r_0 (vibrationally averaged); [b]atomisation energy (two bonds), experiment $0.2231\,E_h$ [6]; n.c.: the 2split runs do not converge (the two sides trade cells and the symmetry breaks). OH: pseudo-O +3 (only the outer 3-shell carried, as 2hom lone pair +1 behind a plane), $r_c = 0.59$, $R = 1.65, 1.83, 2.0$; 2split $R_e \approx 1.69$; dipole 3.6 D (2hom), 2.7 D (2split) against 1.66. [c]Single point at the observed geometry; the binding is the atomisation energy. H_2O: pseudo-O +3, the outer shell as three single electrons in 120° wedges, $r_c = 0.65$; dipole +1.01 D against 1.85, right sign. NH_3: pseudo-N +3 (lone pair frozen, $r_c = 0.81$) and +5 (lone pair a 2hom domain, $r_c = 0.82$); see Section 7. [d]CH_4: 2split, T_d imposed, $R = 1.85$–2.45; energies at 2.054 (Section 6.4). The axisymmetric reduction is used for X_2 and for linear molecules with fixed walls only. Remaining rows to be filled from the runs in progress.

6.1 LiH: pseudo-Li and one bonding pair

LiH is the simplest polar bond with an inner shell, and is viewed as ionic (Li^+H^-) rather than covalent: its dipole, 5.88 D, is 77% of that of point charges ±1 at $3.015\,a_0$ (7.66 D). The model reduction reflects this: the Li $1s^2$ core is excluded by a fixed sphere of radius $r_c = 1.99\,a_0$ around the Li nucleus, which then carries charge +1 (pseudo-Li); the two valence electrons form one 2hom pair over pseudo-Li and a bare proton. The radius is the one at which the pseudo-Li atom alone reproduces the Li ionisation energy; the one-dimensional all-electron Li atom places its own core–valence interface at 1.98–$2.05\,a_0$ over boxes 6–$20\,a_0$, within 3% of it. All runs are `molecule.js`, $h = 0.1$, box $20\,a_0$; binding energies are $E(\mathrm{Li}) + E(\mathrm{H}) - E(\mathrm{LiH})$ with both atoms built the same way and in the same box.

R	E	$D(R)$	μ (D)
2.6	−0.76776	0.0880	—
2.8	−0.77436	0.0946	6.45
3.0	−0.77259	0.0928	6.53
3.2	−0.76709	0.0873	6.66
3.4	−0.75999	0.0802	6.82
exp.		0.092 at 3.015	5.88

A fit through the five points gives $R_e = 2.85\,a_0$ (2.81–2.89 over three-, four- and five-point fits), 5% short of experiment, and $D_e = 0.0947\,E_h$, 3% over. At $R = 3.0$ the dipole is 6.53 D, of the right sign (Li^+H^-) and 11% high; it is 86% of the point-charge value at that R, so the pair, free to distribute itself over both centres, settles largely on H. Nothing was adjusted to LiH. The 2split on the same pseudo-Li (Li valence and H electron in separate domains) gives a slightly longer bond, $R_e = 2.98$–2.99 (−1%), and under-binds, $D_e = 0.063$–0.073 (Section 5.5). The core is a hard exclusion with a radius set by the atom: a frozen-core model, stated as such. A 2hom core held inside the same fixed sphere is equivalent for the valence, since a spherical charge inside the sphere acts outside it as a point charge. The sphere must be fixed: a free boundary between core and valence carries no Pauli cost, and with it the valence presses into the core and $E(R)$ falls monotonically as H approaches.

6.2 BH and BeH_2

BH. Pseudo-B +3 carries its outer 2+3 shell as a 2hom lone pair and one electron, the two separated by a fixed surface through the B nucleus; the B|H boundary is free. Two divisions of the 3 were run, each with its own r_c fitted to the B ionisation energies (box $12\,a_0$, $h = 0.1$): a plane perpendicular to the bond (each domain half the solid angle, $r_c = 1.08$), and a cone about the bond holding one third of the solid angle for the single electron and two thirds for the pair (half-angle 70.5°, $r_c = 1.11$), in proportion to their electrons. For the bond, 2hom (one pair over the B front and a bare proton):

2hom BH	R_e	D_e	μ (D)
plane, $\frac{1}{2}$:$\frac{1}{2}$	2.29	0.155	3.5
cone, $\frac{1}{3}$:$\frac{2}{3}$	2.17	0.224	1.09
exp.	2.329	0.135	1.27

The proportional division corrects the dipole, by moving charge behind B, and costs bond length and binding: the pair confined to the cone draws the proton in. The plane gives bond and energy, the cone the dipole; neither gives all three. 2split binds more strongly in both (0.276 and about 0.34), with dipoles 4.0 and 1.2–1.3 D. Table 1 uses the plane.

BeH_2. Pseudo-Be +2 (2+2, $r_c = 1.37$) carries its two valence electrons as single electrons on either side of a fixed plane through Be, each facing one H. With each bond a 2hom pair over a Be half and a bare proton, $R_e = 2.44$ against the gas-phase $r_0 = 2.52$ (−3%), and the atomisation energy is $0.248\,E_h$ against 0.2231 [6] (+11%). The 2split construction does not settle: the two H domains and the two Be halves trade cells without end, the symmetry breaks (a residual dipole of 0.2 D), and the convergence window never closes.

6.3 Homonuclear X_2

For X_2 the axisymmetric solver is used. Each X is a pseudo-atom whose inner shells, as given by the Atom Simulator configuration, are frozen into a core of charge Z_x with r_c fitted to the sum

of the ionisation energies of the electrons carried; the outer shell lies along the axis as a domain behind X (a 2hom lone pair, or one electron) and one bonding electron, separated by a fixed plane through X. The two bonding electrons form either a 2split pair (the midplane) or one 2hom domain between the nuclei. Only a σ pair is representable. R-scan 0.8–1.3 R_{exp}, $h \approx 0.1$ (adjusted per R to keep the nuclei on cell faces, a jitter of about $\pm 0.005\,E_h$):

X_2	configuration	r_c	R_e: 2hom	/ 2split	D_e: 2hom	/ 2split	exp. R_e / D_e
Li_2	2+1, +1	2.00	5.1	5.15	0.021	0.054	5.05 / 0.038
Na_2	2+4+4+1, +1	2.16	5.5	5.55	0.020	0.050	5.82 / 0.028
B_2	2+2+1, +1	0.88	2.87	2.9	0.045	0.109	3.005 / 0.110
C_2	2+2+2, +2	0.91	2.55	2.58	0.242	0.322	2.348 / 0.232
N_2	2+3+2, +2	0.59	1.95	1.95	0.370	0.480	2.074 / 0.364
O_2	2+3+3, +3	0.59	2.0	2.05	0.349	0.48	2.282 / 0.192
F_2	2+4+3, +3	0.49	no minimum		> 0.35 / > 0.52		2.668 / 0.061

For Li_2, Na_2, B_2, C_2 and N_2, cores +1 and +2, the experimental binding lies between the two constructions and the bond length is within 10%; B_2 and N_2 require the Atom Simulator configurations (as pseudo-+3 with a lone pair, B_2 over-binds twofold). O_2 over-binds about twofold with a short bond, and F_2 has no minimum: its binding grows down to 0.8 R_{exp}. Other O and F configurations do not help (as +2, 2+3+1+2 and 2+3+2+2, both lose the minimum), and a +1 core cannot carry them, since a single electron over a hard core on a unit charge binds by less than 0.5 E_h, below IE_1 of O and F.

Enlarging r_c beyond the ionisation fit moves the error rather than removing it. For O at $r_c = 0.8$ (pseudo-O 16% short of its ionisation energies) the O_2 minimum returns to 2.33 (+2%) with D_e still 0.262 (2hom); OH (3D, 2hom) binds 0.185 at the observed bond (+9%) but its minimum moves beyond 2.0; H_2O (3D, observed geometry) still binds 0.584. For F, $r_c = 1.1$ gives a minimum at 3.07 (+15%) with $D_e = 0.20$, three times experiment. No single r_c serves the oxygen molecules, and none makes F_2 weak. O_2 and F_2 are the molecules whose weakness is attributed, in standard quantum mechanics, to lone-pair (antibonding π^*) repulsion, the Pauli cost that free walls do not charge (Section 7).

Transfer of the carbon model: C_2H_2. The C_2 carbon (pseudo-C +2, $r_c = 0.91$, valence 1+1) carried unchanged into acetylene, with every bond a 2hom pair (each C–H pair the H electron and the outer C half over a bare proton; the C–C pair the two facing halves), has all its walls fixed at the planes through the C nuclei, so the axisymmetric solution is that of the three-dimensional construction. Each bond scanned with the other at its observed length:

C_2H_2	C–C	C–H	atomisation at obs. geometry
2hom	2.50 (+10%)	2.0 (0%)	0.573–0.579
2split*	2.55 (+12%)	1.95 (−3%)	0.896
exp.	2.274	2.009	≈ 0.646

*The 2split C|H walls are free and are held plane, an approximation. The experimental atomisation lies between the two constructions, 2hom the closer (−11%); the C–H bond is right, and the C–C bond is 10% long, as in C_2 (+9%). The carbon model transfers with a consistent C–C bias, plausibly the single σ pair standing in for multiple bonding. Carrying the carbon valence instead as one 2hom pair per atom ($r_c = 0.81$) gives C_2 the right length (+2%) but $D_e = 0.359$ (+55%), a binding of two closed pairs across a free wall that is the missing Pauli repulsion, and is not used.

Other linear molecules. The same pseudo-atoms, unchanged (C +2, N +2 with one electron behind, O +3 with a 2hom lone pair behind; each bond one 2hom σ pair, a fixed plane at every heavy nucleus), give the following linear molecules (axisymmetric, $h \approx 0.1$; R_e from a parabola through the scan about R_{exp}; atomisation at the observed geometry):

molecule	bond	R_e (exp.)	atomisation (exp.)
HCN	C–N	2.25 (2.179), +3%	0.475 (≈ 0.50)
	C–H	1.97 (2.014), −2%	
CO	C–O	2.19 (2.132), +3%	0.344 (0.413)
CO_2	C–O, both	2.15 (2.192), −2%	0.681 (≈ 0.61)
CH	C–H	1.93 (2.116), −9%	0.181 (≈ 0.134)
CH_2, linear	C–H, both	1.97 (2.03, bent)	0.355 (≈ 0.30)

Where carbon sits between two partners on the axis the bonds are within 3%, even with the +3 oxygen of CO and CO_2, whose lone pairs face away from each other, unlike those of O_2. Where carbon ends the chain with an electron behind it (CH, linear CH_2) the C–H bond is short: the remaining carbon electrons of the real radical lie off the axis, which a σ-only model cannot represent.

6.4 CH_4: geometry with an energy offset

Pseudo-C +4 ($1s^2$ frozen, $r_c = 1.0$) carries its four valence electrons as single electrons in four fixed tetrahedral sectors; each H keeps its electron (2split) or joins its sector as a 2hom pair. Three-dimensional, box $12\,a_0$, $h = 0.1$:

R (a_0)	1.85	2.054	2.25	2.45
$E(T_d)$	−11.09131	−11.18844	−11.21482	−11.18861
atomisation	3.73	3.82	3.85	3.82
$E(D_{4h})$, square planar		−11.15120	−11.18467	
$E(D_{4h}) - E(T_d)$		+0.037	+0.030	

The bond is $R_e = 2.25$, 10% long, the same bias as C–C in C_2 and C_2H_2. The atomisation, $3.8\,E_h$ against 0.67, is in error by a nearly constant $3.15\,E_h$: it changes by 0.03 between 2.05 and 2.45, while the part that varies with R fixes the bond. The tetrahedron is preferred to the square plane (four 90° quadrant sectors, the same electrons and r_c) at both lengths, by 0.03–0.04 E_h; in standard quantum chemistry planar CH_4 lies roughly $0.2\,E_h$ above. The shape is right, and the angular stiffness about six times too soft.

The offset belongs to the electron count around carbon. Other ways of carrying the same four electrons give other offsets at $R = 2.054$: two 2hom pairs in half-spaces ($r_c = 0.92$, fitted the same way) 4.21, one four-electron domain ($r_c = 0.87$, outside the pair rule, run as a test) 1.80; so it does not follow the number of walls between carbon's domains. The kinetic energy shows where it comes from. By the virial theorem a bond at equilibrium raises the kinetic energy by D_e; here, with the H atom at $T = 0.454$ on the same mesh, forming CH_4 *lowers* it by 0.84 (four sectors) and 0.86 (two pairs), where +0.67 is required, and the potential energy falls by 3.0–3.4 instead of 1.34. The free walls let the pairs relax into wide, nodeless sector shapes; in the molecule the eight valence electrons of real carbon fill s and p levels, whose nodes cost kinetic energy. The error is the missing Pauli kinetic energy of Section 7, but it acts on the electron count of a centre, not on the walls, and it is carried almost entirely by the composition. It cancels in everything at fixed composition—bond lengths, angles, conformations—and in processes that keep each atom's bonding, and it does not cancel in atomisation or in reactions that change coordination.

6.5 Forces at the observed geometry

The force on a nucleus is read as the pull of the converged density, $\mathbf{F}_a = Z_a \int \rho\,(\mathbf{x}-\mathbf{R}_a)/|\mathbf{x}-\mathbf{R}_a|^3$ outside the kernel clamp, plus the nuclear repulsion. It equals $-\partial E/\partial \mathbf{R}_a$ only if the energy is stationary in everything that moves with the nucleus. A bare proton inside a 2hom bond pair carries no wall of its own, and there the readout agrees with the slope of $E(R)$: CH_4 $+0.011$ and -0.043 against $+0.026$ and -0.049 at $R = 2.25$ and 2.45 (zero at 2.29 against the energy minimum 2.32), LiH, BeH_2 and BH within 0.002–0.011. With 2split the H domain's wall moves with H and is placed by continuity, not by the energy, and the readout misses that wall term (BH: -0.040 against $+0.086$; CH_4: $+0.08$ against 0). Forces are therefore read with bond pairs and bare protons; the spread between equivalent H atoms, about ± 0.005, is the mesh noise at $h = 0.1$.

At the observed geometry the residual force on each H, along the bond (positive: the bond would lengthen) and across the angle (positive: the angle would close), is, in E_h/a_0:

molecule	geometry	along bond	across angle
LiH	3.0	-0.022	
BH	2.3	-0.013	
BeH_2	2.5	-0.018	
OH	1.83	-0.028	
H_2O	1.809, 104.5°	-0.083, -0.096	$+0.009$, $+0.014$
walls released		-0.080	$+0.016$
NH_3	1.913, 106.7°	$+0.086$ to $+0.098$	$\approx +0.035$
CH_4	2.054, T_d	$+0.102$ to $+0.113$	0 (symmetry)
axisymmetric, $-\mathrm{d}E/\mathrm{d}R$ over $\pm 10\%$ of the bond:			
HCN	C–N / C–H	$+0.042$ / -0.013	
CO	C–O	$+0.045$	
CO_2	C–O, per bond	-0.022	
C_2H_2	C–C	$+0.19$	
CH	C–H	-0.046	

For hydrides with cores up to +3 the residual force is 0.013–0.028, a few hundredths of a bohr at typical bond stiffness. The heavy centres with many electrons push their H out (NH_3, CH_4, $+0.1$) or pull it in (H_2O, -0.09). A force must be read against the stiffness: the triple bond of C_2H_2 carries $+0.19$ for a 10% error, CO_2 -0.022 for 2%.

Angles. The angular forces close the angle in both free cases, H_2O and NH_3. For water the energy at fixed O–H = 1.809 falls steadily as the angle closes, with the oxygen wedges fixed and with their walls released after 500 steps (free walls placed by continuity):

H–O–H	95°	104.5°	115°
E, fixed wedges	-5.42394	-5.41638	-5.41240
E, walls released	-5.39597	-5.38444	-5.37126

The minimum lies below 95° in both, so the closing is not set by the sector frame. The two bond pairs meet at a wall on the bisector that charges nothing for pressing them together, and only the proton repulsion holds the angle open; the lone pairs, frozen into a spherical core, push nothing. The angular force at the observed geometry is small (0.01–0.016), but the bend is soft, so the minimum moves far. (Releasing the walls raises the energy by 0.03, since continuity is not energy-minimising, and raises the dipole from 1.01 to 1.43 D against 1.85.) Bond lengths and symmetry-fixed shapes are in the model's range; free angles at a centre with several pairs are not yet.

7 Limitations

Polar bonds. Where the two sides of a bond differ strongly, the true domain boundary curves around the smaller or more weakly bound atom, so heteronuclear molecules are computed in three dimensions only. LiH is recovered once its valence electrons form one bonding pair over an excluded core (Section 6.1); with one electron per domain the dipole remains wrong in sign, so for polar bonds the pair construction, not the boundary shape, is decisive. NaH and HF are still to be run this way.

Inner shells. Pauli exclusion is included through disjoint domains: electrons in different domains do not overlap. A 2hom pair overlays two electrons of opposite spin in one domain, as Pauli permits, and so captures that part of it too. In standard quantum mechanics exclusion is the antisymmetry of the many-electron wave function; imposed as orthogonality of, say, $2s$ to $1s$, it puts a node in the valence orbital inside the core, and the kinetic energy of that node keeps the valence out (the term a pseudopotential replaces). Disjoint domains are orthogonal by construction, $\int u_k u_l = 0$, but by separation rather than by a node: the valence amplitude is free at the wall and pays no node energy, and a free boundary between a shell and the valence, placed by continuity, carries no counterpart of that cost. Inner shells therefore cannot be represented by domains with free walls—the valence presses into them and the bond collapses—and are excluded by a fixed core radius per atom. This is a frozen-core model, and its radius is a property of the atom, set by its ionisation energy.

Strong pseudo-cores. The same missing cost limits the valence a pseudo-atom may carry. With cores of charge up to +3 (LiH, BeH_2, BH, OH, and H_2O with pseudo-O +3) the 2hom binding is within 0.003–0.3 E_h of experiment. With +5 and +6, carrying the whole valence shell, the valence floods the protons at no kinetic price: H_2O with pseudo-O +6 over-binds by 2.5 E_h, NH_3 with pseudo-N +5 by 3 E_h, and CH_4 with pseudo-C +4 ($r_c = 1.0$, four single electrons, no lone pair) by 3.2 E_h, so the cause is the charge of the core, not lone pairs; the kinetic energy rising by less than a third of what the added H electrons alone should cost. Freezing the lone pair instead (pseudo-N +3) keeps the error to 1.4 E_h but leaves a spherical core and inverts the dipole of NH_3. The missing term is the Pauli kinetic energy between domains; until it is supplied, the energies of molecules whose heavy atom carries more than three valence electrons are outside the model's range. Their geometry need not be: in CH_4 the over-binding is a near-constant offset, and the bond and shape survive (Section 6.4). Between two heavy atoms the limit is lower: in X_2 cores up to +2 work, while O_2 and F_2, with a lone pair on each side of the bond, over-bind or collapse (Section 6.3).

The Pauli kinetic energy between domains. The limitations above have one source. The kinetic energy of a density ρ is at least $T_{\mathrm{vW}}[\rho]$, the value it would have if every electron shared one nodeless amplitude $\sqrt{\rho}$; this is exact for one electron and for an opposite-spin pair in one state. Beyond two, Pauli exclusion puts electrons of the same spin into mutually orthogonal states, orthogonality forces nodes, and nodes cost curvature: the excess $T_{\mathrm{P}} = T_s - T_{\mathrm{vW}} \geq 0$. In the uniform electron gas it is the Fermi pressure $\propto \rho^{5/3}$ that keeps matter from collapsing. Between atoms it is the steep short-range repulsion of overlapping closed shells, the resistance of a core to valence (the $2s$ node inside $1s$), and the mutual repulsion of lone pairs. It acts between pairs: two pairs each carry both spins, so wherever two pairs overlap, electrons of equal spin overlap.

RealQM treats the two places differently. Within a domain the kinetic energy is exact, since a domain holds one electron or a 2hom pair and T_{vW} is the whole of it; this is why the pair rule stops at two. Between domains it keeps electrons apart by separation rather than by nodes: at a free wall

the amplitude may stay flat up to the wall, and being kept apart costs only a mild confinement, where overlapping same-spin orbitals would need a node. The model therefore under-charges T_{P} wherever domains are crowded. In an atom the error is largely absorbed: the radial walls fall near the true nodes (for Li the computed core–valence interface is at 1.98–2.05, against $r_c = 1.99$ from the ionisation energy), and r_c is fitted to the ionisation energies. In a molecule the new walls are lateral, between electrons that share the same region, and nothing absorbs the error. With few electrons per centre it is small (LiH, BH, Li_2, C_2, N_2); with many it decides the result: CH_4 with pseudo-C +4 over-binds by $3.2\,E_h$, O_2 twofold, and F_2—whose weak bond is, in standard quantum mechanics, the lone-pair repulsion itself—has no repulsive wall at all. Because the missing term grows as domains are pressed together, it can change geometry as well as energy: the O_2 bond is 10% short. In CH_4 it hardly does: the error there is a near-constant offset attached to the electron count of carbon, and the kinetic energy falls on bonding where the virial theorem requires it to rise (Section 6.4).

Supplying it means charging density that presses on a wall between domains whose electrons would have to be orthogonal. Candidates are a surface energy $\frac{\beta}{2}\int u_k^2$ on such walls (Section 5.3; in axisymmetric tests it did not give bond length and binding together), a node condition on walls between electrons of equal spin, and a local Fermi-gas term in $\rho^{5/3}$ less its von Weizsäcker part, applied where domains meet. A cost on walls alone is not supported by CH_4, whose offset does not follow the number of walls (Section 6.4); and since domains are disjoint, any local functional of the density adds up over them, so a $\rho^{5/3}$ term must be non-local or act at the walls to see the crowding. The test of any of them: F_2 weakly bound with a minimum near $2.67\,a_0$, CH_4 and H_2O no longer over-bound, and LiH, Li_2, C_2, N_2 left where they are.

Large molecules. The last reduction step (reduced valence charges, $h = 0.25$–$0.4\,a_0$) has been checked against reference chemistry only on small fragments: the formamide dimer and a five-residue polyglycine helix, both on 200^3, give bond and hydrogen-bond geometries in line with reference values. Folding runs on whole proteins use contact restraints; without them the fold is not found.

Contacts between molecules. A new domain boundary in a region of high density adds the same uncharged wall as the 2split bond, without a 2hom alternative. Intermolecular energies are outside the model's range, and so is dispersion.

Energies across compositions. Binding and reaction energies compare systems with different domain structures, whose errors differ; only differences within one composition (conformations, scans along a coordinate) are quotable. The offset is attached to the electron count around each atom, so it also cancels in a process that joins fragments without changing any atom's bonding—association, docking, or a reaction written so that every atom keeps its bonds (an isodesmic design, used for the same reason in standard quantum chemistry). Atomisation energies, bond dissociation, and reactions that change coordination do not cancel it. Reaction energies from fragments of unchanged bonding are the intended next step; they are small, of the order of the mesh and step noise at $h = 0.1$, and will need step and mesh ladders.

8 Conclusion

Solved as it stands, the hard-partition model of RealQM carries every electron pair in one of two parameter-free constructions, 2split and 2hom, the limits of complete and of no avoidance; for H_2

and He the exact energy lies between them. A shared pair (core, bonding pair) is 2hom, and for LiH and BH the 2hom bond is the closer to experiment. Bond lengths are good where the domain boundaries are flat or fixed by symmetry. Inner shells must be excluded by a fixed core radius, since a free wall has no Pauli cost; with that, LiH as one 2hom bonding pair has its bond within 5%, its binding within 3% and a dipole of the right sign. Other polar bonds are [**pending**]. Nothing is adjusted to molecular data; each atom contributes one core radius, set from its ionisation energies.

The model is a geometry model. Where an atom carries many valence electrons the missing Pauli kinetic energy shows as a large offset in the binding, nearly independent of geometry and set by the composition; CH_4 over-binds by $3.2\,E_h$ and still has its bond within 10% and the tetrahedron preferred, though with too little angular stiffness. Bond lengths of linear molecules built from the same pseudo-atoms (C_2H_2, HCN, CO, CO_2) are within 3–10%, and the residual forces on H at the observed geometry are 0.01–$0.03\,E_h/a_0$ for cores up to +3. Free angles are not yet right: in H_2O and NH_3 the bond pairs meet at a wall that charges nothing, and the angle closes. Atomisation energies are therefore a diagnostic; energies are quotable within one composition, and for processes that join fragments without changing any atom's bonding.

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